

Copulas and financial applications

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1 Introduction

In finance and insurance, a frequent question: random vectors modeling $\Rightarrow$ [dependence modeling](#)

Example 1. multi-asset derivatives For most assets, there exist T -maturity complex options.

- Worst of options: payoff $V_T = \min(r_{1,T}, r_{2,T})$;
- many structured products;
- basket credit derivatives: payoff
 $V_T = (\sum_{i=1}^m 1(\tau_i \leq T) - K)^+$, τ_i = default time of issuer i ;
- hybrid products, quanto options;
- counterparty risk on derivatives, etc.

$\Rightarrow$ we need the joint law of several asset values at T , or at different intermediate dates.

Example 2. portfolio management

m assets, with returns $(r_{i,t})_{t=1,\dots,T}$, $i = 1, \dots, m$.

Portfolio P&L = $\Delta P_t = P_{t+\Delta t} - P_t = \sum_{i=1}^m N_i r_{i,t} \Delta t$.

Risque measures as value-at-risk or Expected Shortfall:

$$VaR_\alpha : \mathbb{P}(-\Delta P_t \leq VaR_\alpha) = \alpha, \alpha \in (0, 1).$$

Practical problem: what is the law of ΔP_t ?

Example 3. An insurance portfolio, with many different risks: default events, catastrophic risks, mortality, etc. Problem of heterogeneity....

⇒ dependence between some random variables.

Most of the time, one "knows" rather well the marginal laws

Ex.: The Breeden-Litzenberger theorem yields risk-neutral densities

Ex.: individual CDS premia yield the laws of firm default times

Once our margins are "known", what do I need to get the joint law?

Answer: **a copula !**

2 Basics of copulas

A reminder

Consider a map $G : \mathbb{R}^m \rightarrow [0, 1]$. *Notation:* $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{a} \in \mathbb{R}$,

$$(\mathbf{x}_{-k}, \mathbf{a}) := (x_1, \dots, x_{k-1}, \mathbf{a}, x_{k+1}, \dots, x_m).$$

$$\Delta_{k,\mathbf{a}}^b G(\mathbf{x}) := G(\mathbf{x}_{-k}, b) - G(\mathbf{x}_{-k}, \mathbf{a}), \quad k = 1, \dots, m, \quad \mathbf{a}, b \in \mathbb{R}.$$

The G -volume of a "box" $\mathbf{B}_{\mathbf{a},\mathbf{b}} := \prod_{k=1}^m (a_k, b_k]$ is

$$V_G(\mathbf{B}_{\mathbf{a},\mathbf{b}}) = \Delta_{1,\mathbf{a}_1}^{b_1} \cdots \Delta_{m,\mathbf{a}_m}^{b_m} G(\mathbf{x}),$$

that does not depend on $\mathbf{x}$!

The map is m -increasing iff $V_G(\mathbf{B}_{\mathbf{a},\mathbf{b}}) \geq 0$ for every $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}^m$.

A continuous map G is a m -dimensional cdf iff

- 1 it is m -increasing,
- 2 $G(\mathbf{x}) = 0$ when one component of $\mathbf{x}$ is $-\infty$,
- 3 $G(+\infty, \dots, +\infty) = 1$.

Every univariate cdf $F : \mathbb{R} \mapsto \mathbb{R}$ is right-continuous with left-hand limits.

Define the generalized inverse $F^{-}(t) := \inf\{x | F(x) \geq t\}$.

$F \circ F^{-}(u) \geq u$ for every $u \in [0, 1]$, with equality if F is continuous at $F^{-}(u)$.

$F^{-} \circ F(t) \leq t$ for every $t \in \mathbb{R}$

Let X be a univariate random vector.

Lemma 1

If $X \sim F$, then $F(X) \sim \mathcal{U}([0, 1])$ iff F is continuous.

Lemma 2

Assume F is a continuous univariate cdf. Then $F(X) \sim \mathcal{U}([0, 1])$ iff $X \sim F$.

Lemma 3

If $U \sim \mathcal{U}([0, 1])$ then $F^{-}(U) =: X \sim F$ for any univariate cdf F .

Useful for simulation purpose!

Proof (Shorack & Wellner, Th. 1):

$U \leq F(x) \Rightarrow F^{-}(U) \leq x$ (c.f. definition of F^{-}). Inversely, if

$X := F^{-}(U) \leq x$, then $F(x + \varepsilon) \geq U$ for every $\varepsilon > 0$. Since F is right-hand continuous, $F(x) \geq U$. We have proved that $(X \leq x) = (U \leq F(x))$. $\square$

Definition of a copula

Let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$, whose joint cdf is H and its margins are $F_1, \dots, F_m$.

$$\mathbb{P}(X_1 \leq x_1, \dots, X_m \leq x_m) = \mathbb{P}(\mathbf{X} \leq \mathbf{x}) =: H(\mathbf{x}).$$

Definition 1

A m -dimensional copula is a cumulative distribution function $C : [0, 1]^m \rightarrow [0, 1]$ whose margins are uniform.

$m = 2$: $C(u, 1) = u$, $C(1, v) = v$, $C(u, 0) = C(0, v) = 0$ for every $(u, v) \in [0, 1]^2$.

A copula = a dependence function = “what is left after controlling the margins”

Theorem 1 (Sklar)

For any cdf H , with margin F_k , $k \in \{1, \dots, m\}$, there exists a m -dimensional copula C s.t.

$$H(\mathbf{x}) = C(F_1(x_1), \dots, F_m(x_m)), \quad \mathbf{x} \in \mathbb{R}^m, \quad (1)$$

and we can impose that C is continuous.

At the opposite, consider any copula C and m univariate cdfs' F_k , $k = 1, \dots, m$, that satisfy (1). Then, this relationship defines a cdf H in $\mathbb{R}^m$ whose margins are the F_k .

Note that C is **not uniquely defined in general**: C is called **a copula** associated to H .

Sklar's theorem: proof

Consider $\mathbf{X} \sim H$. Let $V_1, \dots, V_m$ be m mutually independent r.v. that are uniform on $[0, 1]$ and that are independent of $\mathbf{X}$.

$$U_k := V_k F_k(X_k) + (1 - V_k) F_k(X_k -).$$

(a). $U_k \sim \mathcal{U}([0, 1])$ for every k .

Indeed, consider any $t \in [0, 1]$. If $t = F_k(x_0)$ for some x_0 , then $(U_k \leq t) = (X_k \leq x_0)$ and $\mathbb{P}(U_k \leq t) = F_k(x_0) = t$.

If t lies between two values of F_k , there exists $\alpha \in (0, 1)$ and x_0 s.t. $t = (1 - \alpha)F_k(x_0 -) + \alpha F_k(x_0)$.

$$\begin{aligned} \mathbb{P}(U_k \leq t) &= \mathbb{P}(X_k < x_0) + \left(\frac{t - F_k(x_0 -)}{F_k(x_0) - F_k(x_0 -)} \right) \mathbb{P}(X_k = x_0) \\ &= \mathbb{P}(X_k < x_0) + t - F_k(x_0 -) = t. \end{aligned}$$

Indeed:

$$\begin{aligned}\mathbb{P}(U_k \leq t) &= \mathbb{P}(U_k \leq t, X_k < x_0) + \mathbb{P}(U_k \leq t, X_k = x_0) \\ &+ \mathbb{P}(U_k \leq t, X_k > x_0) = T_1 + T_2 + T_3\end{aligned}$$

If $X_k < x_0$, then $U_k \leq F_k(X_k) \leq F_k(x_0-) \leq t$. Thus,
 $T_1 = \mathbb{P}(X_k < x_0)$.

$$\begin{aligned}T_2 &= \mathbb{P}(V_k F_k(x_0) + (1 - V_k) F_k(x_0-) \leq t) \mathbb{P}(X_k = x_0) \\ &= \mathbb{P}(V_k (F_k(x_0) - F_k(x_0-)) \leq t - F_k(x_0-)) \mathbb{P}(X_k = x_0) \\ &= \left(\frac{t - F_k(x_0-)}{F_k(x_0) - F_k(x_0-)} \right) \mathbb{P}(X_k = x_0).\end{aligned}$$

If $X_k > x_0$, then $U_k \geq F_k(x_0) \geq t$. Thus, $T_3 \leq \mathbb{P}(U_k = t) = 0$.

(b). Define C as the cdf of $\mathbf{U} := (U_1, \dots, U_m)$ (i.e. a copula). Since its margins are uniform, they are continuous, C itself is continuous: for every $\mathbf{u}$ and $\mathbf{v}$ in $[0, 1]^m$,

$$\begin{aligned} |\mathbb{P}(\mathbf{U} \leq \mathbf{u}) - \mathbb{P}(\mathbf{U} \leq \mathbf{v})| &\leq \sum_{k=1}^m |\mathbb{P}(U_k \leq u_k) - \mathbb{P}(U_k \leq v_k)| \\ &\leq \sum_{k=1}^m |u_k - v_k|. \quad (\text{prove it by induction}) \end{aligned}$$

(c). Note that, for every $\mathbf{x} \in \mathbb{R}^m$,

$$\mathbb{P}(U_1 \leq F_1(x_1), \dots, U_m \leq F_m(x_m)) = \mathbb{P}(\mathbf{X} \leq \mathbf{x}).$$

Indeed, $(U_k \leq F_k(x_k)) = (X_k \leq x_k)$, for every k .

(d). At the opposite, why is H (as defined by (1)) a true cdf?

Easy to check that $H(\mathbf{x}) = 0$ when $x_k = -\infty$ for some k , and

$$H(+\infty, \dots, +\infty) = C(1, \dots, 1) = 1.$$

Moreover, H is m -increasing because it is the case for C . $\square$

Theorem 2 (Sklar V2)

Let H be a continuous cdf in $\mathbb{R}^m$. There exists a unique m -dimensional copula C s.t.

$$H(\mathbf{x}) = C(F_1(x_1), \dots, F_m(x_m)), \quad \mathbf{x} \in \mathbb{R}^m. \quad (2)$$

In such a case, C is called **the copula** of H .

Proof: Since H is continuous, it is the case for the margins F_k and $F_k \circ F_k^-(u) = u$ for every $u \in [0, 1]$. Then, (2) implies

$$C(\mathbf{u}) = H(F_1^-(u_1), \dots, F_m^-(u_m)), \quad \mathbf{u} \in [0, 1]^d.$$

Therefore C is uniquely defined. $\square$

The copula way of working:

$$H \longleftrightarrow (C, F_1, \dots, F_m)$$

Model specification into two parts:

- 1 the margins F_k on one side,
- 2 a copula C on another side.

$\Rightarrow$ a m -dimensional model (law).

Vocabulary: if $\mathbf{X}$ is a random vector in $\mathbb{R}^m$, a (resp. the) copula associated to $\mathbf{X}$ is a (resp. the) copula associated to H , the cdf of $\mathbf{X}$, by Sklar's theorem.

The copula(s) of H are uniquely defined on $\prod_{k=1}^m \text{Im}(F_k)$.

Proposition 1

Let U and V be two r.v. that are uniformly distributed on $[0, 1]$. Let F and G two univariate cdfs'. Set $X := F^{-}(U)$ and $Y := G^{-}(V)$. For every $(x, y) \in \mathbb{R}^2$

$$(X \leq x, Y \leq y) = (U \leq F(x), V \leq G(y)),$$

Denote by $C_{(U,V)}$ the cdf of (U, V) , i.e. their (unique. Why?) copula. This implies

$$\mathbb{P}(X \leq x, Y \leq y) = C_{(U,V)}(U \leq F(x), V \leq G(y)).$$

$\Rightarrow$ the law of (U, V) = a copula of $(X, Y) = (F^{-}(U), G^{-}(V))$, even if F and G are discontinuous.

Proposition 2

Let X and Y be two random variables, $X \sim F$, $Y \sim G$. Assume the cdfs' F and G are continuous. Set $U := F(X)$ and $V := G(Y)$ (that are uniformly distributed on $[0, 1]$). Then, the cdf of (U, V) is the copula of (X, Y) .

The latter properties are valid in any dimension m .

Proposition 3

Copulas are Lipschitz-continuous: for every $(\mathbf{u}, \mathbf{v}) \in \mathbb{R}^{2m}$,

$$|C(u_1, \dots, u_m) - C(v_1, \dots, v_m)| \leq \sum_{k=1}^m |u_k - v_k|.$$

Proposition 4

The components of $\mathbf{X}$ are mutually independent iff a copula of $\mathbf{X}$ is the “independence copula” $C^\perp$:

$$C^\perp(\mathbf{u}) = u_1 \cdots u_m, \mathbf{u} \in [0, 1]^m.$$

Proposition 5

Let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$ whose copula is C , and $\psi_1, \dots, \psi_m$ some strictly increasing maps from $\mathbb{R}$ to $\mathbb{R}$. Then, C is a copula of $(\psi_1(X_1), \dots, \psi_m(X_m))$.

$$\psi_1 \nearrow \text{ and } \psi_2 \searrow: C_{(\psi_1(X_1), \psi_2(X_2))}(u, v) = u - C_{(X_1, X_2)}(u, 1 - v).$$

$$\psi_1 \searrow, \psi_2 \nearrow: C_{(\psi_1(X_1), \psi_2(X_2))}(u, v) = v - C_{(X_1, X_2)}(1 - u, v).$$

$$\psi_1 \searrow, \psi_2 \searrow: C_{(\psi_1(X_1), \psi_2(X_2))}(u, v) = u + v - 1 + C_{(X_1, X_2)}(1 - u, 1 - v).$$

In dimension 2: for every u, v in $[0, 1]$,

$$0 \leq V_{(0,u] \times (v,1]}(C) = C(u, 1) - C(u, v) - C(0, 1) + C(0, v) = u - C(u, v),$$

$$0 \leq V_{(u,1] \times (0,v]}(C) = C(1, v) - C(u, v) - C(1, 0) + C(u, 0) = v - C(u, v),$$

implying $C(u, v) \leq \min(u, v) =: C^+(u, v)$.

C^+ is a copula, the Fréchet upper bound.

$$\begin{aligned} 0 &\leq V_{(u,1] \times (v,1]}(C) = C(1, 1) - C(u, 1) - C(1, v) + C(u, v) \\ &= 1 - u - v + C(u, v), \end{aligned}$$

implying $C(u, v) \geq \max(u + v - 1, 0) =: C^-(u, v)$.

C^- is a copula, the Fréchet lower bound.

Copulas bounds

In dimension m : $C(\mathbf{u}) \leq \min(u_1, \dots, u_m) =: C^+(\mathbf{u})$. C^+ is a copula, the Fréchet upper bound.

$$C(u, v) \geq \max(u_1 + \dots + u_m - m + 1, 0) =: C^-(\mathbf{u}).$$

C^- is a copula **only when $m = 2$** .

Indeed, let $V = (1/2, 1]^m$ be an hypercube in $[0, 1]^m$. Since C^- is symmetrical,

$$\begin{aligned} 0 \leq V_B(C^-) &= C^-(1, \dots, 1) - mC^-(\tfrac{1}{2}, 1, \dots, 1) \\ &+ \frac{m(m-1)}{2} C^-(\tfrac{1}{2}, \tfrac{1}{2}, 1, \dots, 1) + \dots \\ &= 1 - m \max(\tfrac{1}{2} + (m-1) - m + 1, 0) \\ &+ \frac{m(m-1)}{2} \max(\tfrac{1}{2} + \tfrac{1}{2} + (m-2) - m + 1, 0) + 0 = 1 - m/2. \end{aligned}$$

The Fréchet bounds = the extreme level of dependencies between the components of the underlying random vector. Assume X and Y are two real continuous variables (i.e. F and G are continuous).

Definition 2

A subset $A \subset \mathbb{R}^2$ is comonotone (or perfectly positively dependent) if, for every (x_1, y_1) and (x_2, y_2) in A , $(x_1 < x_2) \Rightarrow y_1 \leq y_2$. A

subset $B \subset \mathbb{R}^2$ is counter-monotone (or perfectly negatively dependent) if, for every (x_1, y_1) and (x_2, y_2) in B , $(x_1 < x_2) \Rightarrow y_1 \geq y_2$.

+ a graph

Theorem 3

Let $(X, Y) \sim H$ be a bivariate random vector, H continuous, $X \sim F$, $Y \sim G$. The following proposition are equivalent

- 1 the copula of (X, Y) is C^+ ;
- 2 $\mathbb{P}((X, Y) \in A) = 1$ for some comonotone subset $A \subset \overline{\mathbb{R}}^2$;
- 3 $H(x, y) = \min(F(x), G(y))$;
- 4 $(X, Y) \stackrel{\text{law}}{=} (F^-(U), G^-(U))$ for some $U \sim \mathcal{U}([0, 1])$;
- 5 $(X, Y) = (F^-(G(Y)), G^-(F(X)))$ a.s.;
- 6 X and Y are comonotone.

In particular, if $F = G$ a.s., then $X = Y$ a.s.

Theorem 4

Let $(X, Y) \sim H$ be a bivariate random vector, H continuous, $X \sim F$, $Y \sim G$. The following proposition are equivalent

- 1 the copula of (X, Y) is C^- ;
- 2 $\mathbb{P}((X, Y) \in B) = 1$ for some counter-monotone subset $B \subset \mathbb{R}^2$;
- 3 $H(x, y) = \max(F(x) + G(y) - 1)$;
- 4 $(X, Y) \stackrel{\text{law}}{=} (F^-(U), G^-(1 - U))$ for some $U \sim \mathcal{U}([0, 1])$;
- 5 $(X, Y) = (F^-(1 - G(Y)), G^-(1 - F(X)))$ a.s.;
- 6 X and Y are counter-monotone.

In particular, if X and Y are uniformly distributed on $[0, 1]$, then $X + Y = 1$ a.s.

The concepts of comonotoneity (and the properties above) can be extended in dimension $m > 2$.

The copula of $\mathbf{X}$ is C^+ iff $\mathbf{X} \stackrel{law}{=} (F_1^-(U), \dots, F_m^-(U))$ for some $U \sim \mathcal{U}([0, 1])$, or iff every margin X_k is a strictly increasing map of another margin.

A similar extension is **not possible** for counter-monotoneity.

We have defined a partial order across copulas: for every copula C ,

$$C^- \leq C \leq C^+, \quad C^- \leq C^\perp \leq C^+,$$

and C^- is a copula when $m = 2$.

There exist some totally ordered families.

Ex.: Cuadras-Augé copula family

$$C_\theta(u, v) := \min(u, v)^\theta (uv)^{1-\theta}, \quad \theta \in [0, 1].$$

$C_0 = C^\perp$, $C_1 = C^+$, and, when $\theta \leq \theta'$,

$$\frac{C_\theta(u, v)}{C_{\theta'}(u, v)} = \left(\frac{uv}{\min(u, v)} \right)^{\theta' - \theta} \leq 1.$$

As for every cdf, some copulas may have densities w.r.t. the Lebesgue measure on $[0, 1]^m$.

$$c(\mathbf{u}) := \frac{\partial^m}{\partial u_1 \cdots \partial u_m} C(\mathbf{u}).$$

If H and the F_k have densities h and f_k respectively, then

$$c(\mathbf{u}) := \frac{h(F_1^-(u_1), \dots, F_m^-(u_m))}{\prod_{k=1}^m f_k \circ F_k^-(u_k)}.$$

The simplest way of proving that a map $C : [0, 1]^m \rightarrow [0, 1]$ with uniform margins is a copula:

- calculate $c(\cdot)$ and check it is a nonnegative map.
- Then, $C(\mathbf{u}) = \int_0^{u_1} \cdots \int_0^{u_m} c$ for every $\mathbf{u} \in [0, 1]^m$, because the support of c is included into $[0, 1]^m$.

Since $C(u_1, 1, \dots, 1) = u_1$, then $C(1, \dots, 1) = 1$.

Sometimes, it is more relevant to think (model specification!) in terms of survival functions rather than cdfs'.

Ex.: survival analysis, duration models, credit risk... Can we replace marginal cdfs' F_k by the survival maps $1 - F_k$? Copulas' ideas may be adapted to this new framework $\Rightarrow$ **survival copulas**.

In dimension two: a bivariate cdf $H \equiv (C, F, G)$, $(X, Y) \sim H$.

$$\begin{aligned}\mathbb{P}(X > x, Y > y) &= 1 - F(x) - G(y) + H(x, y) \\ &= (1 - F)(x) + (1 - G)(y) - 1 + C(F(x), G(y)) \\ &= \tilde{C}((1 - F)(x), (1 - G)(y)),\end{aligned}$$

by setting $\tilde{C}(u, v) := u + v - 1 + C(1 - u, 1 - v)$. $\tilde{C}$ is a bivariate copula. It is called **a survival copula** associated with (X, Y) .

More generally, in dim m : H a cdf in $\mathbb{R}^m$, with margins $F_1, \dots, F_m$ and marginal survival functions $S_1, \dots, S_m$. There exists a copula $\tilde{C} : [0, 1]^m \rightarrow [0, 1]$ s.t.

$$\mathbb{P}(X_1 > x_1, \dots, X_m > x_m) = \tilde{C}(S_1(x_1), \dots, S_m(x_m)),$$

for every $\mathbf{x} \in \mathbb{R}^m$. And the latter relationship define a copula $\tilde{C}$ ($\simeq$ Sklar's th.).

Let C be a m -dimensional copula, $\mathbf{U} = (U_1, \dots, U_m) \sim C$. Then:

- $(1 - U_1, \dots, 1 - U_m) \sim \tilde{C}$, a survival copula;
- if C admits a density c , so does $\tilde{C}$; it is given by

$$\tilde{c}(\mathbf{u}) := c(1 - u_1, \dots, 1 - u_m), \mathbf{u} \in (0, 1)^m;$$

-

$$\tilde{C}(\mathbf{u}) = \sum_{j=1}^m (-1)^j \sum_{\substack{A \subset \{1, \dots, m\}, |A|=j \\ (\varepsilon_{A,1}, \dots, \varepsilon_{A,m})}} C(1 - \varepsilon_{A,1}u_1, \dots, 1 - \varepsilon_{A,m}u_m),$$

where $\varepsilon_{A,k} := 1(k \in A)$, $k \in \{1, \dots, m\}$.

3 Families of copulas

Gaussian copulas: the most popular copula

Notations: let Φ (resp. ϕ) the cdf (resp. density) of a $\mathcal{N}(0, 1)$, and Φ_{Σ} denotes the cdf of Gaussian random vector $\mathcal{N}(0_m, \Sigma)$ for some correlation matrix Σ .

Definition 3

For some correlation matrix, the m -dimensional Gaussian copula with parameter Σ is defined as

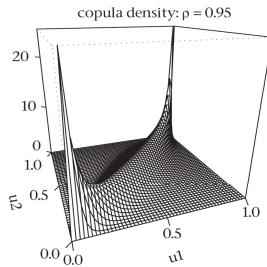
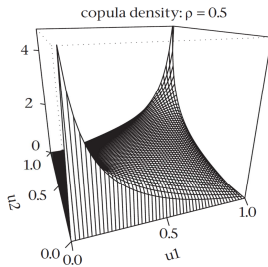
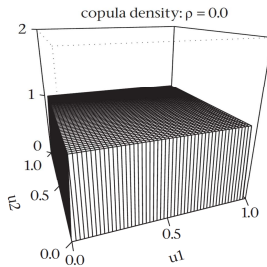
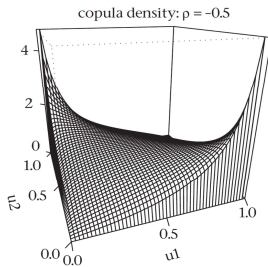
$$C_{\Sigma}(\mathbf{u}) = \Phi_{\Sigma}\left(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_m)\right), \mathbf{u} \in [0, 1]^m.$$

For every $\mathbf{u} \in [0, 1]^m$, set $x_k := \Phi^{-1}(u_k)$, $k \in \{1, \dots, m\}$, and $\vec{x} := (x_1, \dots, x_m)$. The Gaussian copula density is

$$c_{\Sigma}(\mathbf{u}) := \frac{1}{(2\pi)^{m/2} |\Sigma|} \frac{\exp\left(-\vec{x}' \Sigma^{-1} \vec{x} / 2\right)}{\phi(x_1) \dots \phi(x_m)}.$$

$\Sigma =: [\rho_{i,j}]_{1 \leq i,j \leq m}$: a model with $m(m-1)/2$ parameters.
By far, the most widely used copula in finance & insurance.

Plots of bivariate Gaussian copula densities



Gaussian copulas: the most popular copula

Warning: if $\mathbf{X}$ is a Gaussian random vector, its copula is Gaussian.
But the opposite is **false** !!!

Warning: if $\mathbf{X} \sim (C_{\Sigma}, F_1, \dots, F_m)$, then $\rho_{k,l}$ is **not** the correlation between X_k and X_l , nor between U_k and U_l !

Proposition 6

For every $k \in \{1, \dots, m\}$, set $Z_k := \Phi^{-1}(F_k(X_k))$. When the law of X is continuous, $(Z_1, \dots, Z_m) \sim \mathcal{N}(0_m, \Sigma)$.

Therefore, for every (k, l) in $\{1, \dots, m\}$, $\rho_{k,l}$ is the correlation between Z_k and Z_l .

Proof of Prop. 6:

$$\begin{aligned}\mathbb{P}(Z_k \leq z_k, \forall k) &= \mathbb{P}(X_k \leq F_k^- \circ \Phi(z_k), \forall k) \\ &= C_\Sigma(F_k(F_k^- \circ \Phi(z_k)), \forall k) \\ &= C_\Sigma(\Phi(z_1), \dots, \Phi(z_m)) \\ &= \Phi_\Sigma(\Phi^{-1}(\Phi(z_1)), \dots, \Phi^{-1}(\Phi(z_m))) \\ &= \Phi_\Sigma(z_1, \dots, z_m). \quad \square\end{aligned}$$

Gaussian copulas: inference

$\Rightarrow$ a simple inference procedure: an i.i.d. sample $(\mathbf{X}_i)_{i=1,\dots,n}$, $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,n})$.

- 1 replace the unknown margins by the empirical cdfs':

$$F_k(t) \simeq \hat{F}_k(t) := \frac{1}{n} \sum_{i=1}^n 1(X_{i,k} \leq t), k = 1, \dots, m.$$

- 2 set $Z_{i,k} := \Phi^{-1} \circ \hat{F}_k(X_{i,k})$, $i = 1, \dots, n$.
- 3 estimate $\rho_{k,l}$ by the empirical correlation between the Z_k and the Z_l :

$$\hat{\rho}_{k,l} := \frac{1}{\hat{\sigma}_k \hat{\sigma}_l} \sum_{i=1}^n (Z_{i,k} - \bar{Z}_k)(Z_{i,l} - \bar{Z}_l),$$

$$\hat{\sigma}_k^2 := \frac{1}{(n-1)} \sum_{i=1}^n (Z_{i,k} - \bar{Z}_k)^2, k = 1, \dots, m.$$

Gaussian copulas: financial application

Pricing of **basket credit derivatives**: “the” Gaussian copula model

A basket of m defaultable securities (bonds, CDS) + a complex structured product whose payoff is $V_T = \psi(\tau_1, \dots, \tau_m)$ ($\tau_k =$ default time of k)

- 1 the law of τ_k is deduced (on a daily basis) by quotes of several CDS written on k (with different maturities);
- 2 the joint law of $(\tau_1, \dots, \tau_m)$ is obtained assuming a Gaussian copula;
- 3 for standard CDOs, a unique parameter ρ ($\simeq$ implied volatility).

Historical ref.: David Li (2000). On default correlation: a copula function approach.

Extensions towards other copulas: Burtschell, Gregory, Laurent (2008). A comparative analysis of CDO pricing models.

Gaussian copulas: pros and cons

Pros.

- flexible (many parameters)
- ease of inference
- simple to add some components to a random vector $\mathbf{X} \in \mathbb{R}^m$ by keeping a Gaussian copula

Cons.

- no tail dependence (see below)
- symmetrical
- too many parameters in large dimension?

Student copulas: a direct extension of Gaussian copulas.

Notations: $\mathcal{T}_\nu$ = cdf of a univariate centered Student law with ν degrees of freedom. $\mathcal{T}_{\Sigma,\nu}$ = cdf of a centered multivariate Student random vector for some **correlation matrix** Σ , whose density is

$$t_{\Sigma,\nu}(\mathbf{x}) := \frac{\Gamma((\nu + m)/2)}{\Gamma(\nu/2)(\nu\pi)^{m/2}|\Sigma|^{1/2}} \left(1 + \frac{1}{\nu}\mathbf{x}'\Sigma^{-1}\mathbf{x}\right)^{-(\nu+m)/2}.$$

Definition 4

The m -dimensional Student copula with parameter Σ and ν degrees of freedom is defined as

$$C_{\Sigma,\nu}(\mathbf{u}) = \mathcal{T}_{\Sigma,\nu}\left(\mathcal{T}_\nu^{-1}(u_1), \dots, \mathcal{T}_\nu^{-1}(u_m)\right), \mathbf{u} \in [0, 1]^m.$$

$C_{\Sigma,\nu}$ is close to the Gaussian copula C_Σ when ν is “large” One additional parameter that allows fat tails...

Reminder: Elliptical distributions.

Parameterized by $\mu \in \mathbb{R}^m$, a $m \times m$ positive definite matrix Σ and a measurable function $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+ \cup \{+\infty\}$, so that its density w.r.t the Lebesgue measure in $\mathbb{R}^m$ is

$$f_{\mathbf{X}}(\mathbf{x}) = |\Sigma|^{-1/2} g((\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)), \mathbf{x} \in \mathbb{R}^m. \quad (3)$$

Denote $\mathbf{X} \sim \mathcal{E}_m(\mu, \Sigma, g)$, with a cdf $H_{g, \Sigma, m}$.

g = the “density generator”, that must satisfy the constraint

$$\int_0^\infty r^{d-1} g(r^2) dr = \int_0^\infty t^{d/2-1} g(t) dt / 2 = \Gamma(d/2) / (2\pi^{d/2}). \quad (4)$$

Conversely, any nonnegative g function that satisfies (4) can be used as the density generator of an elliptical distribution.

Reminder: Elliptical distributions.

Ex.: $\mathbf{X} \sim \mathcal{N}(\mu, \Sigma)$, by setting $g_{\text{Gauss}}(u) := \exp(-u/2)/(2\pi)^{m/2}$.

Theorem 5 (polar decomposition)

A random vector $\mathbf{X} \sim \mathcal{E}(\mu, \Sigma, g)$ iff $\mathbf{X} \stackrel{\text{law}}{=} \mu + RA^T U$, for some $m \times m$ matrix A s.t. $A^T A = \Sigma$, where U is a random vector uniformly distributed on the m -dimensional unit sphere and R is a nonnegative r.v. independent of U .

When R has a density g_R w.r.t. the Lebesgue measure on $\mathbb{R}^+$ and no mass at zero, this is the case for $\mathbf{X}$ too.

The density $f_{\mathbf{X}}$ of $\mathbf{X}$ is then given by (3) and $g_R(r) = 2\pi^{m/2} r^{m-1} g(r^2)/\Gamma(m/2)$, $r \geq 0$.

Reminder: [Elliptical distributions](#).

Nice properties of elliptical distributions:

- extensions of Gaussian and Student random vectors;
- elliptically contoured iso-density surfaces (symmetrical);
- flexible (parametric/NP density generator);
- stability of sub-vectors, linear transformations and conditional sub-vectors.

Ref.: Cambanis et al. (1981). On the theory of elliptically contoured distributions.

Gomez et al. (2003). A survey on continuous elliptical vector distributions.

Definition 5

A m -dimensional copula C is said to be meta-elliptical if there exists an elliptical distribution in $\mathbb{R}^m$ whose copula is C

The margins of a $\mathcal{E}_m(0, \Sigma, g)$ have the same density

$$f_g(t) = \frac{\pi^{(m-1)/2}}{\Gamma((m-1)/2)} \int_0^{+\infty} g(t^2 + s) s^{(m-3)/2} ds.$$

Set its marginal cdfs' $F_g(x) = \int_{-\infty}^x f_g(t) dt$, and its quantile function $Q_g(u) = \inf\{x; F_g(x) \geq u\}$, $u \in [0, 1]$.

By Sklar's theorem, our meta-elliptical distribution $\mathcal{ME}_d(\Sigma, g)$, whose cdf is C , is given by

$$\begin{aligned} C(\mathbf{u}) &= H_{g, \Sigma, m}(Q_g(u_1), \dots, Q_g(u_m)) \\ &= \int_{-\infty}^{Q_g(u_1)} \cdots \int_{-\infty}^{Q_g(u_m)} g(\mathbf{x}' \Sigma^{-1} \mathbf{x}) d\mathbf{x}. \end{aligned}$$

$\Rightarrow$ the associated meta-elliptical copula density w.r.t. the Lebesgue measure exists and is given by

$$c(\mathbf{u}) = \frac{g(\vec{Q}_g(\mathbf{u})\Sigma^{-1}\vec{Q}_g(\mathbf{u})')}{|\Sigma|^{1/2} \prod_{k=1}^m f_g(Q_g(u_k))}, \text{ a.s.,}$$

$$\vec{Q}_g(\mathbf{u}) := [Q_g(u_1), \dots, Q_g(u_m)].$$

Pros: significantly more flexible than Gaussian and/or Student copulas

Cons: analytical complexity. Still symmetrical.

Let $h : [0, 1] \rightarrow \overline{\mathbb{R}}$, $h(1) = 0$, h is strictly decreasing.

If $h(0) = +\infty$, h will be called a strict generator. Otherwise, define $h^{-1}(u) := 0$ when $u > h(0)$. Notation: $\psi := h^{-1}$.

Define a map $C : [0, 1]^m \rightarrow [0, 1]$ as

$$C(\mathbf{u}) := h^{-1}(h(u_1) + \dots + h(u_m)), \mathbf{u} \in [0, 1]^m.$$

Theorem 6 (Kimberling, 1974)

If h^{-1} is C^∞ and $h(0) = +\infty$, C is a copula *for every m* iff h^{-1} is completely monotone, i.e. $(-1)^p (h^{-1})^{(p)} \geq 0$, $p \in \mathbb{N}$.

Prove that a completely monotone generator is convex and its inverse too!

It can be proved that, when $m = 2$, C is a copula iff h is convex.

Archimedean copulas: examples

Gumbel: $h(u) = (-\ln(u))^\alpha$, for some $\alpha > 1$, and

$$C(\mathbf{u}) := \exp \left(- \left\{ \sum_{k=1}^m (-\ln u_k)^\alpha \right\}^{1/\alpha} \right).$$

Clayton: $h(u) = u^{-\alpha} - 1$ for some $\alpha > 0$, and

$$C(\mathbf{u}) := \left(\sum_{k=1}^m u_k^{-\alpha} - m + 1 \right)^{-1/\alpha}.$$

Frank: $h(u) = -\ln \left((\exp(-\alpha u) - 1) / (\exp(-\alpha) - 1) \right)$, $\alpha > 0$ if $m \geq 3$, and

$$C(\mathbf{u}) := \frac{(-1)}{\alpha} \ln \left\{ 1 + \frac{1}{(\exp(-\alpha) - 1)^{m-1}} \prod_{k=1}^m (\exp(-\alpha u_k) - 1) \right\}.$$

AMH: $h(u) = \ln \left((1 - \theta)/u + \theta \right)$, $\theta \in [0, 1)$.

Plots of some bivariate Archimedean copula

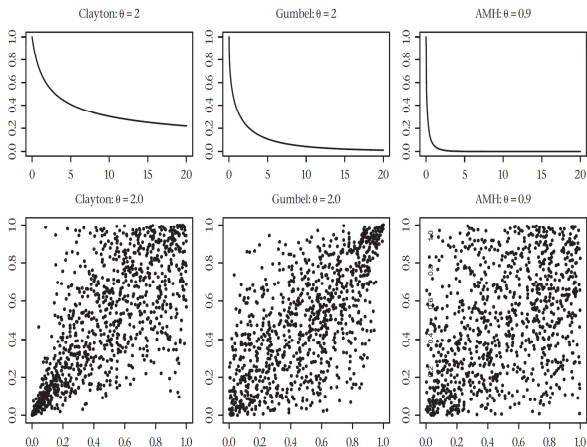


Figure: Scatter plots from several bivariate Archimedean copulas and their respective “generators” $\psi = h^{-1}$

Theorem 7

A map $\psi : \mathbb{R}^+ \mapsto \mathbb{R}^+$ is completely monotone with $\psi(0) = 1$ iff ψ is the Laplace transform of a nonnegative r.v.

$\Rightarrow$ we can generate Archimedean copulas by inverting Laplace transforms.

Ex.: if $X \sim \text{Exp}(\mu)$, then $\psi(t) = \mathbb{E}[\exp(-tX)] = 1/(1 + \mu t)$.
 $h(x) = \psi^{-1}(x) = 1/(\mu x) - 1/\mu$.

The associated Archimedean copula is Clayton(1):

$$C(\mathbf{u}) = \left\{ \mu \left(\sum_{k=1}^m \frac{1}{\mu u_k} - \frac{1}{\mu} \right) + 1 \right\}^{-1} = \left\{ \sum_{k=1}^m \frac{1}{u_k} - m + 1 \right\}^{-1}.$$

Note that many generators can induce the same copula.

Ex.: assume the r.v. X is Gamma distributed, with density

$$f_X(x) = \exp(-x)x^{1/\theta-1}/\Gamma(1/\theta)1(x > 0), \theta > 0.$$

Then $\mathbb{E}[X] = \text{Var}(X) = 1/\theta$,

$$h^{-1}(t) = \psi(t) = \mathbb{E}[\exp(-tX)] = (1+t)^{-1/\theta}.$$

$\Rightarrow h(x) = x^{-\theta} - 1$, and we get a Clayton of parameter θ .

Archimedean copulas: Proof of Theorem 7

($\Leftarrow$): if $\psi(x) = \mathbb{E}[\exp(-xZ)]$, then

$$(-1)^p \psi^{(p)}(x) = \mathbb{E}[Z^p \exp(-xZ)] \geq 0.$$

$\Rightarrow$): for any $a > 0$, let the map $x \mapsto \psi(a - ax)$ from $]0, 1[$ to $\mathbb{R}$.

$$\psi(a - ax) = \sum_{n \geq 0} (-1)^n a^n \frac{\psi^{(n)}(a)}{n!} x^n \geq 0.$$

$$\psi_a(\lambda) := \psi(a - a \exp(-\lambda/a)) = \sum_{n \geq 0} (-a)^n \frac{\psi^{(n)}(a)}{n!} \exp\left(-\frac{n\lambda}{a}\right)$$

ψ_a is the Laplace transform of a discrete distribution μ_a :

$$\mu_a(n/a) = (-a)^n \frac{\psi^{(n)}(a)}{n!}, \quad n \in \mathbb{N}, \quad 0 \text{ elsewhere.}$$

μ_a is a probability: $\sum_{n \geq 0} (-a)^n \psi^{(n)}(a)/n! = \psi(0) = 1$.

Moreover, $\psi_a(\lambda) \rightarrow \psi(\lambda)$ when $a \rightarrow +\infty$:

$$\psi\left(a - a \exp\left(-\frac{\lambda}{a}\right)\right) \simeq \psi\left(a - a\left(1 - \frac{\lambda}{a} + O\left(\frac{1}{a^2}\right)\right)\right)$$

Lemma 4

Let (ω_n) be a sequence of Laplace transforms that is associated with a sequence of measures μ_n , i.e.

$\omega_n(\lambda) = \int_0^{+\infty} \exp(-\lambda x) \mu_n(dx)$, and $\omega_n(\lambda) \rightarrow \omega_\infty(\lambda)$ for every $\lambda > 0$. Then, there exists a measure μ_∞ s.t. $\mu_n \xrightarrow{\text{law}} \mu_\infty$ and ω_∞ is the Laplace transform of μ_∞ .

Apply: there exists μ_∞ s.t. with $\psi(\lambda) = \int_0^{+\infty} \exp(-\lambda x) \mu_\infty(dx)$, $\lambda > 0$. And μ_∞ is a probability since this is the case for the μ_a . $\square$

Lemma 5

A map $g : \mathbb{R}^+ \rightarrow \mathbb{R}$ is completely monotone iff: $\forall m \geq 1, \forall (u, \Delta)$
 s.t. $0 \leq u - m\Delta < \dots < u - \Delta < u < +\infty$,

$$\sum_{k=0}^m (-1)^{m-k} C_m^k g(u - k\Delta) \geq 0. \quad (5)$$

Sketch of the proof: ($\Rightarrow$). From Th. 7, there exists a cdf G s.t.

$$g(u - k\Delta) = \int_0^{+\infty} \exp(-(u - k\Delta)x) G(dx).$$

$$\begin{aligned} \sum_{k=0}^m (-1)^{m-k} C_m^k g(u - k\Delta) &= \int_0^{\infty} \sum_{k=0}^m (-1)^{m-k} C_m^k \exp(k\Delta x) \exp(-ux) G(dx) \\ &= \int_0^{\infty} \left(\exp(k\Delta) - 1 \right)^m \exp(-ux) G(dx) \geq 0. \end{aligned}$$

$\Leftarrow$). C.f. numerical approximation of derivatives and $\Delta \rightarrow 0$. $\square$

Archimedean copulas: proof of Kimberling (1974) theorem

Proof of Th. 6: $\Rightarrow$. Assume that C is an Archimedean copula in $\mathbb{R}^m$ for every m , with a generator h . Let a box $\mathbf{B}_{\mathbf{a},\mathbf{b}} := (\mathbf{a}, \mathbf{b}] \subset [0, 1]^m$, $\mathbf{a} := (a, \dots, a)$, $\mathbf{b} := (b, \dots, b)$, for arbitrary (a, b) .

$$\begin{aligned} V_C(\mathbf{B}_{\mathbf{a},\mathbf{b}}) &= \sum_{k=0}^m (-1)^k C_m^k h^{-1}(kh(a) + (m-k)h(b)) \\ &= \sum_{k=0}^m (-1)^k C_m^k h^{-1}(mh(a) + (m-k)(h(b) - h(a))) \\ &= \sum_{k=0}^m (-1)^{m-k} C_m^{m-k} h^{-1}(mh(a) + k(h(b) - h(a))). \end{aligned}$$

Fix u and Δ , positive numbers. Set a s.t. $mh(a) = u$ and b s.t. $h(a) - h(b) = \Delta < u/m$.

Then (5) is satisfied. Since m is arbitrary, ψ is completely monotone.

($\Leftarrow$. Let $\psi = h^{-1}$ be a completely monotone map and

$$C(\mathbf{u}) := \psi(\psi^{-1}(u_1) + \dots + \psi^{-1}(u_m)).$$

Check easily that the margins of C are uniform.

Since $\psi' \leq 0$, $h' \leq 0$. Moreover,

$$\frac{\partial^m}{\partial_1 \dots \partial_m} C(\mathbf{u}) = \psi^{(m)}(h(u_1) + \dots + h(u_m)) \prod_{k=1}^m h'(u_k) \geq 0.$$

By integration, C is m -increasing and it then a copula, for any m .

□

We have seen that h is convex iff $\psi = h^{-1}$ is convex, and this is equivalent to $C(u, v) = h^{-1}(h(u) + h(v))$ is a copula.

Not all convex generator h (or its inverse) generate Archimedean copulas for every m .

Ex.: if $\psi(u) = \max(1 - u, 0)$, the $h = \psi$ and

$$C(\mathbf{u}) = \max(u_1 + \dots + u_m - m + 1, 0) = C^-(\mathbf{u}),$$

that is not a copula when $m \geq 3$.

Definition 6

A map $f : \overline{\mathbb{R}} \rightarrow \mathbb{R}$ is m -monotone on $[a, b] \subset \overline{\mathbb{R}}^2$, $m \geq 2$, if

- f is $m - 2$ differentiable and $(-1)^k f^{(k)}(x) \geq 0$,
 $k \in \{0, \dots, m - 2\}$, $x \in [a, b]$, and
- $(-1)^{m-2} f^{(m-2)}$ is decreasing and convex on $[a, b]$.

Theorem 8 (McNeil & Neslehova, 2009)

Let h be a generator. For a given m ,

$C(\mathbf{u}) = h^{-1}(h(u_1) + \dots + h(u_m))$ is a m -dimensional copula iff h^{-1} is m -monotone on $[0, +\infty[$.

Corollary 1

Let h be a generator that is m -differentiable, for a given m . Then,

$C(\mathbf{u}) = h^{-1}(h(u_1) + \dots + h(u_m))$ is a m -dimensional copula iff $(-1)^k (h^{-1})^{(k)} \geq 0$, $k \in \{1, \dots, m\}$.

Proposition 7

If f and g are completely monotone (c.m.) , then this is the case for fg . If f is c.m. and $g \geq 0$ with a c.m. derivative, then $f \circ g$ is c.m.

$\Rightarrow$ a way of building new Archimedean families.

Proposition 8

If C is Archimedean with a strict generator, then $C \geq C^\perp$.

Proof of Prop. 8: Let h be the generator of C . There exists a r.v. X s.t., for every nonnegative λ, μ

$$\begin{aligned} h^{-1}(\lambda + \mu) &= \mathbb{E}[\exp(-(\lambda + \mu)X)] \\ &\geq \mathbb{E}[\exp(-\lambda X)] \cdot \mathbb{E}[\exp(-\mu X)] \text{ (c.f. Lemma 6)} \\ &\geq h^{-1}(\lambda) h^{-1}(\mu). \end{aligned}$$

Thus, $h^{-1}(h(x) + h(y)) \geq xy$, for all (x, y) . $\square$.

Lemma 6

(concentration inequality) Let X be a r.v.; f and g are two increasing (or two decreasing) functions. Then,
$$\mathbb{E}[f(X)]\mathbb{E}[g(X)] \leq \mathbb{E}[fg(X)].$$

Proof of Lemma 6: for every (x, y) ,
$$[f(x) - f(y)] \cdot [g(x) - g(y)] \geq 0$$

Let X and Y be two independent r.v. , $X \stackrel{\text{law}}{=} Y$.

$$\begin{aligned}\mathbb{E}[fg(X)] + \mathbb{E}[fg(Y)] &\geq \mathbb{E}[f(X)g(Y)] + \mathbb{E}[f(Y)g(X)] \\ &= \mathbb{E}[f(X)]\mathbb{E}[g(Y)] + \mathbb{E}[f(Y)]\mathbb{E}[g(X)]\end{aligned}$$

$$\Rightarrow 2\mathbb{E}[fg(X)] \geq 2\mathbb{E}[f(X)]\mathbb{E}[g(X)]. \quad \square$$

Definition 7

A m -dimensional copula is associative if, for every $\mathbf{u} \in [0, 1]^{2m-1}$,

$$\begin{aligned} C(C(u_1, \dots, u_m), u_{m+1}, \dots, u_{2m-1}) \\ &= C(u_1, C(u_2, \dots, u_{m+1}), u_{m+2}, \dots, u_{2m-1}) \\ &= \dots = C(u_1, \dots, u_{m-1}, C(u_m, \dots, u_{2m-1})). \end{aligned}$$

If $m = 2$: $C(C(u, v), w) = C(u, C(v, w))$, $\forall (u, v, w) \in [0, 1]^3$.

Theorem 9 (Stupnanova & Kolisarova, 2011)

Let $C : [0, 1]^m \rightarrow [0, 1]$ be a copula. The two assertions are equivalent

- (a) C is associative and $C(u, \dots, u) < u$ when $u \in (0, 1)$.
- (b) C is an Archimedean copula.

$\Rightarrow$ some Goodness-of-fit tests for “Archimedeanity”

Pros.

- simplicity, ease to simulate
- nice theoretical properties
- inference (see below)

Cons.

- symmetrical copulas
- the dependence between any subvector is Archimedean with the same generator
- an insufficient number of parameters

An extension of Archimedean copulas

An attempt, to induce asymmetries among the components of the copula (Liebscher, 2008).

Any Archimedean copula may be written

$$C(\mathbf{u}) = h^{-1}(h(u_1) + \dots + h(u_m)) = \bar{h}^{-1}\left(\prod_{k=1}^m \bar{h}(u_k)\right),$$

with $\bar{h}(t) := \exp(-h(t))$.

For some map $\xi : [0, 1] \mapsto [0, 1]$, $\xi(0) = 0$, $\xi(1) = 1$, define

$$\bar{C}(\mathbf{u}) := \xi\left(\frac{1}{N} \sum_{j=1}^N \prod_{k=1}^m h_{j,k}(\xi^{-1}(u_k))\right),$$

where the maps $h_{j,k} : [0, 1] \mapsto [0, 1]$ are increasing, differentiable, $h_{j,k}(0) = 0$, $h_{j,k}(1) = 1$.

Note that we recover any Archimedean copula with $N = 1$ and $h_{j,k}(t) = t$ for every (j, k) .

Theorem 10

Assume ξ is m -times differentiable, $\xi' > 0$, $\xi^{(i)} \geq 0$ when $i \in \{2, \dots, m\}$. If $N^{-1} \sum_{j=1}^N h_{j,k}(t) = t$ for every $t \in [0, 1]$ and every $k \in \{1, \dots, m\}$ then $\overline{C}$ is a copula.

Ex.: $h_{j,k}(t) = t^{\delta_{j,k}}$, $j < N$ and $h_{N,k}(t) = Nt - \sum_{j=1}^{N-1} t^{\delta_{j,k}}$,
 $\delta_{j,k} \in [1, 2]$, $\sum_{j=1}^{N-1} \delta_{j,k} \leq N$.

Let m increasing maps $g_k : [0, 1] \rightarrow [0, 1]$. Define

$$C(\mathbf{u}) := \prod_{k=1}^m g_k(u_{(k)}), \quad u_{(1)} \leq u_{(2)} \leq \dots \leq u_{(m)}. \quad (6)$$

C is a copula if

- $g_1(u) = u$ for all u ;
- $g_k(1) = 1$ for every k ;
- for every k, j , $k + j \leq m$, for all u, v , $u < v$,

$$\sum_{i=1}^j (-1)^i C_j^i \prod_{l=1}^i g_{k+l}(u) \prod_{l=i+1}^j g_{k+l}(v) \geq 0.$$

Example: $g_i(u) := u^{a_i}$ for every $i = 1, \dots, m$.

Set $\Delta a_k = a_k - a_{k-1}$, $a_k = 0$ when $k \leq 0$,

$$\Delta^2 a_k := \Delta(\Delta a_k) = a_k - 2a_{k-1} + a_{k-2}, \text{ etc.}$$

Proposition 9

In (6), C is a copula iff the sequence $(a_1, \dots, a_m)$ is completely monotone, i.e. iff $(-1)^j \Delta^j a_k \geq 0$ for every k, j .

Ref.: Mai & Scherer (2009). Lévy-frailty copulas.

For reliability & credit models: ease of generating event clusterings, high levels of dependencies.

Illustration: an industrial system, with two outputs 1 and 2.

Default time for the disruption of k 's production = X_k , $k = 1, 2$.

The production lines are partly common:

- a first common process, with a failure time $\tau_{1,2}$,
 $\tau_{1,2} \sim \mathcal{E}(\lambda_{1,2})$, and
- two specific factories, with idiosyncratic failure times τ_k ,
 $\tau_k \sim \mathcal{E}(\lambda_k)$, $k = 1, 2$.

$(\tau_1, \tau_2, \tau_{1,2})$ are mutually independent.

$X_1 = \min(\tau_1, \tau_{1,2})$, $X_2 = \min(\tau_2, \tau_{1,2})$.

What is the copula of (X_1, X_2) ?

$$S_k(x_k) := \mathbb{P}(X_k > x_k) = \mathbb{P}(\tau_k > x_k)\mathbb{P}(\tau_{1,2} > x_k) = \exp(-(\lambda_k + \lambda_{1,2})x_k).$$

$$\begin{aligned}\mathbb{P}(X_1 > x_1, X_2 > x_2) &= \mathbb{P}(\tau_1 > x_1)\mathbb{P}(\tau_2 > x_2)\mathbb{P}(\tau_{1,2} > \max(x_1, x_2)) \\ &= \exp(-\lambda_1 x_1 - \lambda_2 x_2 - \lambda_{1,2} \max(x_1, x_2)) \\ &= \exp(-(\lambda_1 + \lambda_{1,2})x_1 - (\lambda_2 + \lambda_{1,2})x_2 \\ &\quad - \lambda_{1,2} \{\max(x_1, x_2) - x_1 - x_2\}) \\ &= S_1(x_1)S_2(x_2) \exp(\lambda_{1,2} \min(x_1, x_2)) \\ &= S_1(x_1)S_2(x_2) \min(\exp(\lambda_{1,2}x_1), \exp(\lambda_{1,2}x_2)) \\ &= S_1(x_1)S_2(x_2) \min(S_1(x_1)^{-\alpha_1}, S_2(x_2)^{-\alpha_2}),\end{aligned}$$

with $\alpha_k = \lambda_{1,2}/(\lambda_k + \lambda_{1,2})$, $k = 1, 2$.

$\Rightarrow$ the survival copula of (X_1, X_2) is a Marshall-Olkin copula

$$C(u, v) := \min(u^{1-\alpha_1}v, uv^{1-\alpha_2}), \quad \alpha_1, \alpha_2 \in [0, 1].$$

Exponential laws are the unique laws that satisfy the “no memory” property: $\mathbb{P}(X > x + t | X > t) = \mathbb{P}(X > x)$, for every $x, t \geq 0$.

Marshall & Olkin (JASA, 1967) searched for a bivariate law with a similar property: does there exist a random vector (X_1, X_2) s.t.

$$\mathbb{P}(X_1 > x_1 + t, X_2 > x_2 + t | X_1 > t, X_2 > t) = \mathbb{P}(X_1 > x_1, X_2 > x_2),$$

for every nonnegative real numbers x_1, x_2, t ? They impose that the laws of X_1 and X_2 are exponential.

The single solution is the “bivariate exponential” distribution:

$$\mathbb{P}(X_1 > x_1, X_2 > x_2) = \exp(-\alpha_1 x_1 - \alpha_2 x_2 - \alpha_{1,2} \max(x_1, x_2)),$$

i.e. a Marshall-Olkin copula + exponential margins.

Proof: Set $\mathbb{P}(X_1 > x_1, X_2 > x_2) =: S(x_1, x_2)$. We assume

$$S(x_1 + t, x_2 + t) = S(x_1, x_2)S(t, t), \quad S(x_1, 0) = \exp(-\theta_1 x_1), \\ S(0, x_2) = \exp(-\theta_2 x_2), \quad \text{for some } \theta_k > 0, \quad k = 1, 2.$$

Set $x_1 = x_2 =: x$: $S(x + t, x + t) = S(x, x)S(t, t)$, and then $S(x, x) = \exp(-\theta x)$, for some $\theta > 0$.

Choosing $x_2 = 0$:

$$S(x_1 + t, t) = S(x_1, 0)S(t, t) = S(x_1, 0) \exp(-\theta t) = \exp(-\theta_1 x_1 - \theta t).$$

Similarly, $S(t, x_2 + t) = \exp(-\theta_2 x_2 - \theta t)$.

Set $\alpha_k := \theta - \theta_k$, $k = 1, 2$ and $\alpha_{1,2} := \theta_1 + \theta_2 - \theta$:

$$S(x_1, x_2) = \exp \left(-\alpha_1 x_1 - \alpha_2 x_2 - \alpha_{1,2} \max(x_1, x_2) \right). \quad \square$$

In larger dimensions $m > 2$, similar models can be built:

- m outputs, with lifetimes X_k , $k \in \{1, \dots, m\}$;
- for (potentially) any subset $I \subset \{1, \dots, m\}$, a lifetime Z_I , related to the shocks that can affect the outputs indexed in I .
- all the Z_I are mutually independent and exponentially distributed;
- $X_k = \inf_{\mathcal{S}(k)} Z_I$, where $\mathcal{S}(k)$ is the set of subsets I that contain k .

At most $2^m - 1$ “shocks” that induce the default of all or only some of the outputs.

Useful in practice:

- introduction of meaningful clusters (country/industry/...), catastrophic risks;
- potentially high levels of dependencies (joint default events; c.f. “fatal” vs “non fatal” shock models in particular);
- but analytical complexity & difficult inference.

Consider an i.i.d. sample of bivariate random vectors $(X_i, Y_i)_{i=1, \dots, n}$ with marginal cdfs' F and G and a copula C .

Set $X_{(n)} := \max_i X_i$, and $Y_{(n)} := \max_i Y_i$.

$$\mathbb{P}(X_{(n)} \leq x) = \mathbb{P}(X_i \leq x, \forall i) = F^n(x).$$

Similarly, $\mathbb{P}(Y_{(n)} \leq y) = G^n(y)$.

$$\begin{aligned}\mathbb{P}(X_{(n)} \leq x, Y_{(n)} \leq y) &= \mathbb{P}(X_i \leq x, Y_i \leq y, \forall i) \\ &= \prod_{i=1}^n \mathbb{P}(X_i \leq x, Y_i \leq y) \\ &= C(F(x), G(y))^n = C\left((F^n(x))^{1/n}, (G^n(y))^{1/n}\right)^n.\end{aligned}$$

$\Rightarrow$ the copula $C_{(n)}$ of $(X_{(n)}, Y_{(n)})$ is $C_{(n)}(u, v) := C^n(u^{1/n}, v^{1/n})$.

Ex.: if $C = C^\perp$ then $C_{(n)} = C^\perp$.

Ex.: if $C = C^+$, then $C_{(n)}^+(u, v) = \left(\min(u^{1/n}, v^{1/n}) \right)^n = C^+(u, v)$.

Ex.: if $C = C^-$, then

$$C_{(n)}^-(u, v) = \left(\max(u^{1/n} + v^{1/n} - 1, 0) \right)^n \neq C^-(u, v),$$

but $C_{(n)}^-$ is really a copula!

Definition 8

A copula $C : [0, 1]^m \rightarrow [0, 1]$ is max-stable if, for every $r > 0$, every $\mathbf{u} \in [0, 1]^m$,

$$C(\mathbf{u}) = C^r(u_1^{1/r}, \dots, u_m^{1/r}).$$

Definition 9

A copula $C_* : [0, 1]^m \rightarrow [0, 1]$ is said “extreme-value”, if there exists a copula C s.t., for every $\mathbf{u} \in [0, 1]^m$,

$$C_*(\mathbf{u}) = \lim_n C^n(u_1^{1/n}, \dots, u_m^{1/n}).$$

One says that C belongs to the domain of attraction of C_* .

Proposition 10

A copula is “extreme-value” iff it is max-stable.

Proof. $\Leftarrow$). Set $C = C_*$.

$(\Rightarrow$). For every $r > 0$,

$$\begin{aligned} C_*^r(u_1^{1/r}, \dots, u_m^{1/r}) &= \lim_n C^{rn}(u_1^{1/rn}, \dots, u_m^{1/rn}) \\ &= \lim_p C^p(u_1^{1/p}, \dots, u_m^{1/p}) \text{ (continuity of } C) \\ &= C_*(\mathbf{u}). \quad \square \end{aligned}$$

Theorem 11

*A 2-dimensional copula C is max-stable iff there exists a convex map $A : [0, 1] \rightarrow [1/2, 1]$, $A(0) = A(1) = 1$ s.t.
 $\max(t, 1 - t) \leq A(t) \leq 1$ for every $t \in [0, 1]$, and*

$$C(u, v) = \exp \left(\ln(uv) A \left(\frac{\ln v}{\ln(uv)} \right) \right).$$

A is called the Pickands function.

Sketch of the proof of Theorem (11) [See Joe, 1997]:

($\Leftarrow$ comes from

$$C(u^{1/r}, v^{1/r})^r = \exp\left(\frac{\ln(uv)}{r} A\left(\frac{\ln v}{\ln(uv)}\right)\right)^r = C(u, v).$$

and check that C is a copula.

$\Rightarrow$). Set $A(t) := -\ln C(\exp(-(1-t)), \exp(-t))$.

$$\begin{aligned} A\left(\frac{\ln v}{\ln(uv)}\right) &= -\ln C(u^{-1/\ln(uv)}, v^{-1/\ln(uv)}) \\ &= -\ln C(u, v)^{-1/\ln(uv)} = \ln C(u, v) / \ln(uv), \end{aligned}$$

since C is extreme-value. Moreover,

$$C(\exp(-(1-t)), \exp(-t)) \leq C(1, \exp(-t)) = \exp(-t).$$

Then, $A(t) \geq t$ by taking the logarithm. Similarly, $A(t) \geq 1-t$. But the proof of A 's convexity is not obvious. $\square$

Extreme-value copulas

Ex. : Galambos copula

$$C(\mathbf{u}) = \exp \left(- \left(\sum_{k=1}^m (-\ln u_k)^\theta \right)^{1/\theta} \right).$$

Many other examples: bivariate Marshall-Olkin, Hüsler-Reiss, etc.

Proposition 11

Let C be a bivariate extreme-value copula, $(U, V) \sim C$. Define the map $K : [0, 1] \rightarrow [0, 1]$, $K(t) := \mathbb{P}(C(U, V) \leq t)$. Then, $K(t) = t - (1 - \tau)t \ln t$, where τ is the Kendall's tau of C :

$$\tau := 4\mathbb{E}[C(U, V)] - 1.$$

K is called the **Kendall function** of C (or of (U, V)).

Conjecture of Ghoudi, Khoudraji and Rivest (1998): the latter property characterizes extreme-value copulas. Unfortunately, this is false (c.f. Neslehova and Du, 2013).

Similar results can be stated in dimensions larger than 2, but this becomes heavier.

Reminder: if X is univariate and there exist some sequences (a_n) and (b_n) s.t. $(X_{(n)} - a_n)/b_n$ is weakly convergent, then the limiting law is Generalized Extreme Value (GEV), whose cdf is

$$H(x, \gamma) := \exp \left(- \{1 + \gamma x\}_+^{-1/\gamma} \right), \quad x \in \mathbb{R}, \quad \gamma \in \mathbb{R}.$$

- $\gamma = 0$: Gumbel law $H(x, 0) = \exp(-\exp(-x))$ on the whole line;
- $\gamma > 0$: Fréchet law, whose support is $[-1/\gamma, +\infty)$;
- $\gamma < 0$: Weibull law, with the support $(-\infty, -1/\gamma]$.

In dimension m ,

$$\mathbb{P}\left(\frac{X_{(n),1} - a_{1,n}}{b_{1,n}} \leq x_1, \dots, \frac{X_{(n),m} - a_{m,n}}{b_{m,n}} \leq x_m\right) \xrightarrow{\text{law}} \\ C(H(x_1, \gamma_1), \dots, H(x_m, \gamma_m)),$$

for some parameters $\gamma_1, \dots, \gamma_m$, and C is max-stable (extreme-value).

Theorem 12

Let C be a m -dimensional copula. C is max-stable iff

$$C(\mathbf{u}) = \exp \left(- \int_{S_m} \max_{1 \leq i \leq m} \left(\frac{t_i}{\ln(1/u_i)} \right) \mu(d\mathbf{t}) \right),$$

where

- $S_m = \{\mathbf{x} \in \mathbb{R}^m; \sum_{k=1}^m x_k = 1, x_k \geq 0\}$ is the simplex in $\mathbb{R}^m$;
- μ is a borelian measure on S_m (called the spectral measure), that satisfies $\int_{S_m} t_k \mu(d\mathbf{t}) = 1$ for every $k \in \{1, \dots, m\}$.

Ref. Gudendorf and Segers (2009). Extreme-value copulas.

Set $\ell(\mathbf{x}) = \ell(x_1, \dots, x_m) := \int_{S_m} \max_{1 \leq i \leq m} (t_i x_i) \mu(d\mathbf{t})$, for every $\mathbf{x} \in \mathbb{R}_+^m$. Let $\tilde{C}$ be a copula in the domain of attraction of C .

It can be shown that

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{1}{t} \{1 - \tilde{C}(1 - tx_1, \dots, 1 - tx_m)\} \\ = \ell(\mathbf{x}) = -\ln C(\exp(-x_1), \dots, \exp(-x_m)). \end{aligned}$$

With ℓ , we can try to state of Pickands representation theorem in dimensions larger than 2.

Extreme-value copulas in dimension $m > 2$

Let the map $A : S_m \rightarrow [1/m, 1]$ defined as

$$\ell(\mathbf{x}) = (x_1 + \dots + x_m) A\left(\frac{x_1}{\sum_k x_k}, \dots, \frac{x_m}{\sum_k x_k}\right).$$

Every max-stable/extreme-value copula in $\mathbb{R}^m$ can be written

$$C(\mathbf{u}) = \exp\left(\left(\sum_{k=1}^m \ln u_k\right) A\left(\frac{\ln u_1}{\sum_k \ln u_k}, \dots, \frac{\ln u_m}{\sum_k \ln u_k}\right)\right)$$

The map A is convex and satisfy

$$\max_{1 \leq i \leq m} w_i \leq A(\mathbf{w}) \leq 1, \quad \forall \mathbf{w} \in S_m,$$

but the latter property does not characterizes the convenient A maps, except when $m = 2$.

4 Generation of copula families

The simplest way of generating new copula families.

If $C_1, \dots, C_p$ are p copulas in $\mathbb{R}^m$ and $\omega_1, \dots, \omega_p$ are weights in $[0, 1]$, $\sum_{j=1}^p \omega_j = 1$, then

$$\sum_{j=1}^p \omega_j C_j \text{ is a copula.}$$

More generally: let $(C_\theta)_{\theta \in \Theta}$, $\Theta \in \mathbb{R}^q$, be a family of copulas.

Then $C(\mathbf{u}) := \int C_\theta(\mathbf{u}) \mu(d\theta)$ is a copula, for any probability μ .

- Mixture of Gaussian copulas:
 $C(\mathbf{u}) = \omega C_{\Sigma_1}(\mathbf{u}) + (1 - \omega) C_{\Sigma_2}(\mathbf{u});$
- Mixture with rotated copulas: for a given parametric copula family $\{C_\theta; \theta \in \Theta\}$ (Clayton, Gumbel, Gaussian, etc), set

$$C(u, v) = \omega_1 C_{\theta_1}(u, v) + \omega_2 C_{\theta_2, R}(u, v) + \omega_3 C_{\theta_3, \bar{R}}(u, v),$$

$$C_{\theta, R}(u, v) := u - C_\theta(u, 1 - v), \quad C_{\theta, \bar{R}}(u, v) := v - C_\theta(1 - u, v),$$

and $\omega_1 + \omega_2 + \omega_3 = 1$.

- Mixtures of copulas coming from several “usual” parametric families.

Warning: identifiability ! Sometimes, necessary to impose additional constraints, as $\omega < 1/2$...

Theorem 13

Let $\gamma : [0, 1] \rightarrow [0, 1]$ be an increasing continuous map, $\gamma(0) = 0$, $\gamma(1) = 1$. Let $C_\gamma : [0, 1]^2 \rightarrow [0, 1]$ defined as

$$C_\gamma(u, v) := \gamma^{-1}\left(C(\gamma(u), \gamma(v))\right), \quad (u, v) \in [0, 1]^2.$$

The following assertions are equivalent:

- (i) for every bivariate copula C , C_γ is a copula;
- (ii) γ^{-1} is convex.
- (iii) γ is concave;

γ is called a “distortion function”

Proof: (i) $\Rightarrow$ (ii) Let $(a, d) \in [0, 1]^2$, $a \leq d$.

Choose $C = C^- : (u, v) \mapsto \max(u + v - 1, 0)$.

$$u_1 = v_1 = \gamma^{-1}((a+1)/2), \quad u_2 = v_2 = \gamma^{-1}((d+1)/2)$$

$$\Rightarrow \gamma(u_1) = \gamma(v_1) = (a+1)/2, \quad \gamma(u_2) = \gamma(v_2) = (d+1)/2.$$

$$C^-(\gamma(u_1), \gamma(v_1)) = \max((a+1)/2 + (a+1)/2 - 1, 0) = a,$$

$$C^-(\gamma(u_1), \gamma(v_2)) = \max((a+1)/2 + (d+1)/2 - 1, 0) = (a+d)/2,$$

$$C^-(\gamma(u_2), \gamma(v_1)) = (a+d)/2, \quad C^-(\gamma(u_2), \gamma(v_2)) = d.$$

$$C_\gamma^- \text{ is a copula} \Rightarrow \gamma^{-1}(a) - 2\gamma^{-1}((a+d)/2) + \gamma^{-1}(d) \geq 0.$$

$$\Rightarrow \gamma^{-1}\left(\frac{a+d}{2}\right) \leq \frac{1}{2}\{\gamma^{-1}(a) + \gamma^{-1}(d)\}.$$

This implies that γ^{-1} is convex (exercise!).

(i) $\Leftrightarrow$ (ii) Assume γ^{-1} is convex and C is a copula, $(U, V) \sim C$.

$$C_\gamma(0, v) = \gamma^{-1}\left(C(\gamma(0), \gamma(v))\right) = \gamma^{-1}(0) = 0$$

$$C_\gamma(1, v) = \gamma^{-1}\left(C(\gamma(1), \gamma(v))\right) = \gamma^{-1}(\gamma(v)) = v$$

For any $B := (u_1, u_2] \times (v_1, v_2]$, let us prove that $V_C(B) \geq 0$.

Since C is a copula,

$$\begin{aligned} & C(\gamma(u_1), \gamma(v_1)) - C(\gamma(u_1), \gamma(v_2)) - C(\gamma(u_2), \gamma(v_1)) \\ & + C(\gamma(u_2), \gamma(v_2)) =: a - b - c + d \geq 0. \end{aligned}$$

Note that b and c belong to $[a, d] \Rightarrow$ two cases: $a \leq b \leq c \leq d$ or $a \leq c \leq b \leq d$.

It is sufficient to prove

$$\gamma^{-1}(a) - \gamma^{-1}(b) - \gamma^{-1}(c) + \gamma^{-1}(d) \geq 0 \quad (7)$$

$$\begin{aligned} b - a &= \mathbb{P}(U \leq \gamma(u_1), V \in (\gamma(v_1), \gamma(v_2)]) \\ &\leq d - c = \mathbb{P}(U \leq \gamma(u_2), V \in (\gamma(v_1), \gamma(v_2)]). \end{aligned}$$

(7) is deduced from

$$\frac{\gamma^{-1}(b) - \gamma^{-1}(a)}{b - a} \leq \frac{\gamma^{-1}(d) - \gamma^{-1}(c)}{d - c}. \quad (8)$$

Indeed, (8) implies

$$\gamma^{-1}(b) - \gamma^{-1}(a) \leq \left(\frac{b - a}{d - c} \right) \{ \gamma^{-1}(d) - \gamma^{-1}(c) \} \leq \gamma^{-1}(d) - \gamma^{-1}(c). \quad \square$$

Ok if $a \leq b \leq c \leq d$ (γ^{-1} is convex).

If $a \leq c \leq b \leq d$: the maps $x \mapsto (\gamma^{-1}(x) - \gamma^{-1}(a))/(x - a)$ and $x \mapsto (\gamma^{-1}(d) - \gamma^{-1}(x))/(d - x)$ are increasing since γ^{-1} convex.

$$\Rightarrow \frac{\gamma^{-1}(b) - \gamma^{-1}(a)}{b - a} \leq \frac{\gamma^{-1}(d) - \gamma^{-1}(a)}{d - a}, \text{ and}$$

$$\frac{\gamma^{-1}(d) - \gamma^{-1}(a)}{d - a} \leq \frac{\gamma^{-1}(d) - \gamma^{-1}(c)}{d - c}. \quad \square$$

Ex.: $\gamma(t) = t^{1/r}$, $r > 0$ and $C_\gamma(u, v) = C^r(u^{1/r}, v^{1/r})$.

Distortion transforms: generalization in dim m

Similar results in dimension $m \geq 3$.

$$C_\gamma(\mathbf{u}) = \gamma^{-1}\left(C(\gamma(u_1), \dots, \gamma(u_m))\right),$$

where $\gamma : [0, 1] \mapsto [0, 1]$ is increasing, $\gamma(0) = 0$, $\gamma(1) = 1$.

Theorem 14

C_γ is copula for every m -dimensional copula C iff γ^{-1} is absolutely monotone, ie iff $(\gamma^{-1})^{(k)} \geq 0$ for every $k \in \{1, \dots, m\}$.

Ex.: $\gamma(t) = (1 - \exp(-\theta t)) / (1 - \exp(-\theta))$.

Proposition 12

If γ_1 and γ_2 are two distortion functions s.t. γ_1^{-1} and γ_2^{-1} are absolutely monotone, then $\gamma_1 \circ \gamma_2$ is a distortion map and $(\gamma_1 \circ \gamma_2)^{-1}$ is absolutely monotone.

Example: set $\gamma(t) = \Phi(\Phi^{-1}(t) + a)$, for some $a \geq 0$ ([Wang's transform](#)).

It can be proved that γ is distortion map whose inverse is absolutely monotone: $\gamma^{-1}(u) = \Phi(\Phi^{-1}(u) - a)$.

Assume C is a gaussian copula parameterized by the correlation matrix Σ . Then

$$C(\mathbf{u}) := \Phi \left(\Phi^{-1} \left(\Phi_{\Sigma} \left(\Phi^{-1}(u_1) + a, \dots, \Phi^{-1}(u_m) + a \right) \right) - a \right)$$

is a m -dimensional copula.

Ref.: Morillas (2005), Valdez and Xiao (2011)

Let C_1 and C_2 be two bivariate copulas.

Khoudraji (1995) showed that

$$C(u, v) := C_1(u^\alpha, v^\beta)C_2(u^{1-\alpha}, v^{1-\beta}), \quad 0 \leq \alpha, \beta \leq 1,$$

is a copula.

$\Rightarrow$ possibility of mixing features of several copulas, of creating asymmetric copulas, etc.

Extension in dimension m : let $C_1, \dots, C_p$ be p copulas in $\mathbb{R}^m$.

$g_{j,k} : [0, 1] \mapsto [0, 1]$, $j \in \{1, \dots, p\}$, $k \in \{1, \dots, m\}$, be increasing (or equal to the constant map one).

$$\text{Assumption : } \prod_{j=1}^p g_{j,k}(v) = v, \forall v \in [0, 1], \forall k.$$

$$\Rightarrow C(\mathbf{u}) := \prod_{j=1}^p C_j(g_{j,1}(u_1), \dots, g_{j,m}(u_m))$$

is a copula.

Ref;: Liebscher (2008). J. Multivariate Analysis.

Example: for every (j, k) , $g_{j,k}(v) := v^{\theta_{j,k}}$, for some $\theta_{j,k} \in [0, 1]$, and $\sum_{j=1}^p \theta_{j,k} = 1$.

In particular, this yields the copula

$$C(\mathbf{u}) := C_1(u_1^{1-\theta_1}, \dots, u_m^{1-\theta_m}) \prod_{k=1}^m u_k^{\theta_k}, \mathbf{u} \in [0, 1]^m.$$

Example: for every (j, k) , $g_{j,k}(v) := v^{\theta_{j,k}} \exp(\alpha_{j,k}(v - 1))$, $\sum_{j=1}^p \theta_{j,k} = 1$, $\theta_{j,k} \in [0, 1]$, $\sum_{j=1}^p \alpha_{j,k} = 0$, $\alpha_{j,k} \in (-\infty, 1)$, $\theta_{j,k} \geq -\alpha_{j,k}$.

If X is univariate, $X \sim F$, the general simulation procedure along F is:

- draw $U \sim \mathcal{U}([0, 1])$;
- set $X := F^{-}(U)$.

Let (X, Y) be a bivariate random vector, $(X, Y) \sim (C, F, G)$. How can we draw realizations of (X, Y) ?

- 1 independently draw $W_k \sim \mathcal{U}([0, 1])$, $k = 1, 2$.
- 2 set $U = W_1$ and $V = \partial_1 C^{-1}(W_1, \cdot)|_{W_2}$, where $\partial_1 C^{-1}(w_1, \cdot)$ is the inverse map of

$$v \mapsto \frac{\partial}{\partial u} C(w_1, v).$$

- 3 $(X, Y) := (F^{-}(U), G^{-}(V))$.

Proposition 13

The conditional law of V given $U = u$ has the cdf

$$v \mapsto \frac{\partial}{\partial u} C(u, v).$$

And similarly by exchanging the roles of U and V .

Proof:

$$\begin{aligned} \mathbb{P}(V \leq v | U = u) &= \lim_{\Delta u \rightarrow 0} \frac{\mathbb{P}(U \in [u, u + \Delta u], V \leq v)}{\mathbb{P}(U \in [u, u + \Delta u])} \\ &= \lim_{\Delta u \rightarrow 0} \frac{\{C(u + \Delta u, v) - C(u, v)\}}{\Delta u} \\ &= \frac{\partial}{\partial u} C(u, v) \cdot \square \end{aligned}$$

A similar result in dimension $m \geq 3$: we want some realizations of $\mathbf{X} \sim (C, F_1, \dots, F_m)$, where C is a m -dimensional copula.

Definition 10

The Rosenblatt transform associated to C is the map $\mathcal{R} : [0, 1]^m \rightarrow [0, 1]^m$ defined as follows: if $\mathcal{R}(\mathbf{u}) = \mathbf{v}$ then $v_1 = u_1$ and

$$v_k = \frac{\frac{\partial^{k-1}}{\partial u_1 \dots \partial u_{k-1}} C(u_1, \dots, u_k, 1, \dots, 1)}{\frac{\partial^{k-1}}{\partial u_1 \dots \partial u_{k-1}} C(u_1, \dots, u_{k-1}, 1, \dots, 1)}, \quad k \in \{2, \dots, m\}.$$

Proposition 14

$\mathbf{U} \sim C$ if and only if $\mathcal{R}(\mathbf{U}) \sim C^\perp$.

An almost universal way of simulating along random vectors:

- 1 independently draw $W_k \sim \mathcal{U}([0, 1])$, $k = 1, \dots, m$.
- 2 set $\mathbf{U} = \mathcal{R}^{-1}(W_1, \dots, W_m)$.
- 3 $\mathbf{X} := (F_1^{-1}(U_1), \dots, F_m^{-1}(U_m))$.

Ok, but the inversion of $\mathcal{R}$ may be analytically demanding!

$\Rightarrow$ in practice, search for smarter simulation procedures, well-suited for your particular model.

Archimedean copula case: $\mathbf{U} \sim C$, C is the Archimedean copula

$$C : \mathbf{u} \mapsto h^{-1}\left(\sum_{k=1}^m h(u_k)\right),$$

and h^{-1} is the Laplace transform of a positive r.v. Y (see Th. 7).

- 1 independently draw $W_k \sim \mathcal{U}([0, 1])$, $k = 1, \dots, m$.
- 2 independently draw Y .
- 3 set $U_k := h^{-1}(-\ln W_k/Y)$, $k \in \{1, \dots, m\}$.

$$\begin{aligned}\mathbb{P}(\mathbf{U} \leq \mathbf{u}) &= \mathbb{P}\left(h^{-1}\left(-\frac{\ln W_k}{Y}\right) \leq u_k, \forall k\right) \\&= \mathbb{P}\left(-\frac{\ln W_k}{Y} \geq h(u_k), \forall k\right) \\&= \mathbb{P}\left(W_k \leq \exp(-Yh(u_k)), \forall k\right) \\&= \mathbb{E}\left[\mathbb{E}\left[1(W_k \leq \exp(-Yh(u_k))), \forall k \mid Y\right]\right] \\&= \mathbb{E}\left[\prod_{k=1}^m \mathbb{E}\left[1(W_k \leq \exp(-Yh(u_k))) \mid Y\right]\right] \\&= \mathbb{E}\left[\exp\left(-Y \sum_{k=1}^m h(u_k)\right)\right] \\&= h^{-1}\left(\sum_{k=1}^m h(u_k)\right) \square.\end{aligned}$$

If $\mathbf{U} \sim C_{\Sigma}$, a **Gaussian copula**: use the properties of Gaussian random vectors and Proposition 6.

Assume $\mathbf{X} \sim (C_{\Sigma}, F_1, \dots, F_m)$.

Reminder: for every $k \in \{1, \dots, m\}$, set $Z_k := \Phi^{-1}(F_k(X_k))$.

When the law of X is continuous, $\mathbf{Z} := (Z_1, \dots, Z_m) \sim \mathcal{N}(0_m, \Sigma)$.

- 1 independently draw $W_k \sim \mathcal{N}(0, 1)$, $k = 1, \dots, m$.
- 2 draw $\mathbf{Z}$ from $\mathbf{W} := (W_1, \dots, W_m)$ (Cholevski decomposition).
- 3 set $X_k := F_k^{-1} \circ \Phi(Z_k)$, $k \in \{1, \dots, m\}$.

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